Macrophyte survey form

Physical variables. Follow the instructions in the section, **Recording the physical variables** in 131_07 Surveying freshwater macrophytes in rivers

River	Till	Site	U/S (control)
Driver (s)	Macrophyte Restr.	Site ID	
Surveyor(s)	RB / MB	Date	09 AUG 2022
*NGR top		Start time	14.30
*NGR bottom		Finish time	1600
*NGR mid-point			

* Record the National Grid Reference (NGR) at the survey length's top and bottom so the survey length can be relocated. Enter the NGR from the survey length's mid-point into BIOSYS.

% wadeable	95%
Total % cover of macrophytes	
Total % cover of filamentous algae (not included in total above)	60

Photo number or reference	upstream	downstream

Typicality	A
Comparable survey lengths	
Comparability of survey lengths	

A <25% of data on this form compromised
B 25-50% of data on this form compromised
C >50% of data on this form compromised

List which survey's this surveys will be compared with. (UWWT only)

Required only for UWWTD surveys
I >75%, II 50-75%, III <50%

Record the percentage of the survey area falling into each of the following categories:

	Width (m) %
<1	
1-5	30
5-10	70
10-20	
>20	

	Depth (m) %
<0.25	60
0.25-0.5	20
0.5-1	10
>1	10

	Substrate %
Bedrock	—
Boulders/cobbles	10
Pebbles/gravel	75
Sand	10
Silt/clay	5
Peat	—

	Bed stability %
Solid/firm	
Stable	
Unstable	100
Soft/sinking	

	Habitats %
Pool	15
Run	60
Riffle	15
Slack	10

	Shading % (left bank)
None	100
Broken	
Dense	

	Shading % (right bank)
None	100
Broken	
Dense	

	Clarity %
Clear	100
Turbid	
Cloudy	

Notes/comments:

Macrophyte Taxa list and Total Cover Values (TCV's). Follow the instructions in the sections, **Recording the macrophytes** and **Calculating the taxon cover values** in 131_07 Surveying freshwater macrophytes in rivers.

What to record

Record **all** macrophytes that you find in the 100m survey length and identify them to species level.

Estimate the percentage cover for each taxon that you record, then assign a Taxon Cover Value (TCV) based on the percentage of the channel area that the macrophyte covers. Use values shown in the table below:

$\bar{x} = 6m$

TCV	Percentage cover for the macrophyte species	
C1	<0.1%	< 0.6
C2	0.1 to 1%	$0.6 - 6m$
C3	1 to 2.5%	$6 - 15m$
C4	2.5 to 5%	15 30
C5	5 to 10%	30 60
C6	10 to 25%	60 150
C7	25 to 50%	150 300
C8	50 to 75%	300 450
C9	>75%	$> 450 - 600$

Notes and actions

The table below describes the special features, notes and actions in the taxa list:

Item in table	Description
Taxa labelled **sp*** or **sp(p)**	Make every effort to identify these taxa to species level, either in the field or back in the laboratory. You may need to confirm the identification with colleagues or Nigel Holmes. Make a record for each unknown species and label them 'species x: 1', 'species x: 2', and so on. You will need to enter each different species as a separate record (with its TCV) on BIOSYS.
Specimen required	These are taxa that that are particularly difficult to identify to species level. Record the site, date, name of recorder, possible identity and supporting notes. You may need to have your identification verified by an independent expert. You may preserve the specimen in your reference collection.
Rare	Taxa which are likely to be found in less than 1% of surveys.
Bold taxa	Taxa from the original Mean Trophic Rank (MTR) manual are shown in bold. Use BIOSYS to give an MTR score in the interim and information for the Water Framework Directive (WFD), as we move to one method and an extended taxa list for all survey purposes.

Taxa (synonym)	TCV	Notes and actions	Taxa (synonym)	TCV	Notes and actions
Macroalgae					
batrachospermum sp(p)			nitella sp*		specimens required
chara sp* (such as vulgaris)			nostoc sp*		specimens required
cladophora aegagropila		specimens required	blue-green algal film / pelts (phormidium / lyngbya sp(p)		
cladophora glomerata, rhizoclonium hieroglyphicum, **blanketweed agg**			**enteromorpha** intestinalis (ulva flexuosa / intestinalis)		
unidentified unbranched filamentous non-slimy green algae excluding cladophora, vaucheria and hydrodictyon			**vaucheria sp(p)**		
hildenbrandia rivularis			zygnematalean (slimy, unbranched, fil algae)		
hydrodictyon reticulatum					
lemanea sp(p) (such as **fluviatilis**)					
Bryophytes	NONE				
amblystegium fluviatile (hygroamblystegium fluviatile)			**hygrohypnum ochraceum**		
amblystegium (hygroamblystegium) tenax			**hyocomium armoricum**		
blindia acuta			**jungermannia sp***		specimens required
brachythecium plumosum			leptodyctium riparium (**amblystegium riparium**)		
brachythecium rivulare			**marsupella sp* emarginata**		specimens required
brachythecium rutabulum			**nardia sp* compressa**		specimens required
bryum pseudotriquetrum			octodiceras fontanum		
calliergon cuspidatum			orthotrichum rivulare		
chiloscyphus polyanthos			palustriella commutata (cratoneuron commutatum)		
cinclidotus fontinaloides			**pellia endiviifolia**		
cratoneuron filicinum			**pellia epiphylla**		
dichodontium flavescens/pellucidum			**philonotis sp* (such as fontana)**		specimens required
dichodontium palustris (**dicranella palustris**)			platyhypnidium riparioides (**rhynchostegium riparioides**)		
fissidens sp* (such as rufulus/crassipes)		specimens required	**polytrichum commune**		
fontinalis antipyretica			**racomitrium aciculare**		
fontinalis squamosa					
hygrohypnum luridum					

Taxa (synonym)	TCV	Notes and actions	Taxa (synonym)	TCV	Notes and actions
riccardia sp* chamaedryfoliamultifida		specimens required	schistidium rivulare		
riccia fluitans			**sphagnum sp*** (such as denticulatum)		specimens required
scapania sp* (such as **undulata**)		specimens required	**thamnobryum alopecurum**		
Lichens					
dermatocarpon sp(p)*		specimens required			
Vascular plants					
azolla filiculoides			**eleocharis palustris**	1	
equisetum fluviatile			**eleogiton fluitans**		
equisetum palustre			**elodea canadensis**		
acorus calamus			**elodea nuttallii**		
alisma lanceolatum			glyceria fluitans agg		includes any small glyceria (not maxima)
alisma plantago-aquatica			**glyceria maxima**	3	
apium inundatum			**groenlandia densa**		
apium nodiflorum			**hippuris vulgaris**		
berula erecta			hottonia palustris		Rare
bidens cernua			**hydrocharis morsus-ranae**		Rare
bidens tripartita			hydrocotyle ranunculoides		Rare
bolboschoenus maritimus		Rare	hypericum elodes		Rare
butomus umbellatus			**iris pseudacorus**		
callitriche hamulata			juncus articulatus		
callitriche hermaphroditica/truncata		specimens required	**juncus bulbosus**		
callitriche obtusangula			**lemna gibba**		
callitriche stagnalis/platycarpa/obtusangula	1	specimens required	**lemna minor**	1	
caltha palustris			**lemna minuta**		
carex acuta	1		**lemna trisulca**		
carex acutiformis	1		**littorella uniflora**		
carex aquatilis			lobelia dortmanna		
carex elata			**lotus pedunculatus**		
carex paniculata			lythrum salicaria		
carex riparia			mentha sp*	-1	specimens required
carex rostrata			mentha aquatica		
carex vesicaria			**menyanthes trifoliata**		
catabrosa aquatica			mimulus sp/hybrid		specimens required
ceratophyllum demersum			**montia fontana**		
cicuta virosa		Rare			

Taxa (synonym)	TCV	Notes and actions	Taxa (synonym)	TCV	Notes and actions
myosotis sp(p)	1	acceptable if non-flowering	**potamogeton natans**		
myosotis laxa			potamogeton nodosus		
myosotis scorpoides	1		**potamogeton obtusifolius**		Rare
myosoton aquaticum			**potamogeton pectinatus**		
myriophyllum alterniflorum			**potamogeton perfoliatus**		
myriophyllum spicatum			**potamogeton polygonifolius**		
nuphar lutea			**potamogeton praelongus**		
nymphaea alba			**potamogeton pusillus**		
nymphoides peltata			**potamogeton trichoides**		Rare
oenanthe aquatica			potamogeton x olivaceus		
oenanthe crocata	1		potamogeton x salicifolius		
oenanthe fistulosa			potentilla palustris		
oenanthe fluviatilis			**potentilla erecta**		
persicaria amphibia			**ranunculus sp (r. sect batrachian sp** or **hybrid indet)**		do not record: ▪ circinatus ▪ omiophyllus ▪ hederaceus
persicaria hydropiper			**ranunculus aquatilis**		record only if plant displays classic characteristics
phalaris arundinacea	1		ranunculus aquatilis var. diffusus (**r.trichophyllus**)		
phragmites australis			**ranunculus circinatus**		
potamogeton alpinus			**ranunculus flammula**		
potamogeton berchtoldii	1		**ranunculus fluitans**		record only if plant displays classic characteristics
potamogeton crispus			**ranunculus hederaceus**		
potamogeton filiformis		Rare	**ranunculus omiophyllus**		
potamogeton friesii		Rare	**ranunculus penicillatus subsp. penicillatus**		record only if plant displays classic characteristics
potamogeton gramineus					
potamogeton lucens					

Taxa (synonym)	TCV	Notes and actions	Taxa (synonym)	TCV	Notes and actions
ranunculus penicillatus subsp. pseudofluitans var pseudofluitans		record only if plant displays classic characteristics	**spirodela polyrhiza**		
ranunculus penicillatus subsp pseudofluitans var vertumnus		record only if plant displays classic characteristics	triglochin palustris		
ranunculus sceleratus			**typha angustifolia**		
rorippa amphibia			**typha latifolia**		
rorippa nasturtium-aquaticum agg.			utricularia sp(p)		Rare specimens required
rorippa palustris			**veronica anagallis-aquatica**		
rumex hydrolapathum			veronica beccabunga		
sagittaria sagittifolia			**veronica catenata**		
schoenoplectus lacustris			**veronica catenata x anagallis-aquatica (or parents indet)**		
scirpus sylvaticus			**veronica scutellata**		
sium latifolium		generally rare, although common western Ireland and fen WFD water bodies	**viola palustris**		
sparganium angustifolium			**zannichellia palustris**		
sparganium emersum					
sparganium erectum					

Additional species

If you go further in your identification than the category on the list above (such as for *spirogyra*) then you must enter the category on BIOSYS on the list that it comes from (such as *zygnematalean*).

Species	Comments
Conocephalum conicum	Ranunculus repens 1
Marchantia polymorpha	Sagina
Lunularia cruciata	Scrophularia
Plagiomnium	Stachys palustris
Rhizomnium	Solanum dulcamara
Achillea millefolium	Trifolium pratense
Agrostis stolonifera 2	Trifolium repens
Chryosplenium oppositifolium	Tussilago farfara
Epilobium hirsutum	Urtica dioica
Equisetum arvense	Valeriana officinalis
Filipendula ulmaria	Sagina procumbens 1
Juncus effusus 1	INNS - Fallopia japonica
Juncus inflexus	INNS - Heracleum mantegazzianum
Lycopus europeaus	INNS - Impatiens glandulifera
Lysimachia vulgaris	INNS – Lyschiton americanus
Montia fontana	
Petasites	

Addition comments on the survey

Standard sketch map for surveys of river macrophytes

Sketch Map. Follow the instructions in the section, **Drawing a sketch map** in 131_07 Surveying freshwater macrophytes in rivers .

River	Till	NGR top	NU 07659 19279
Site	U/S Haeburn	NGR bottom	NU 07650 19370
Date	09 AUG 2023	NGR mid-point	

alt 73m

